



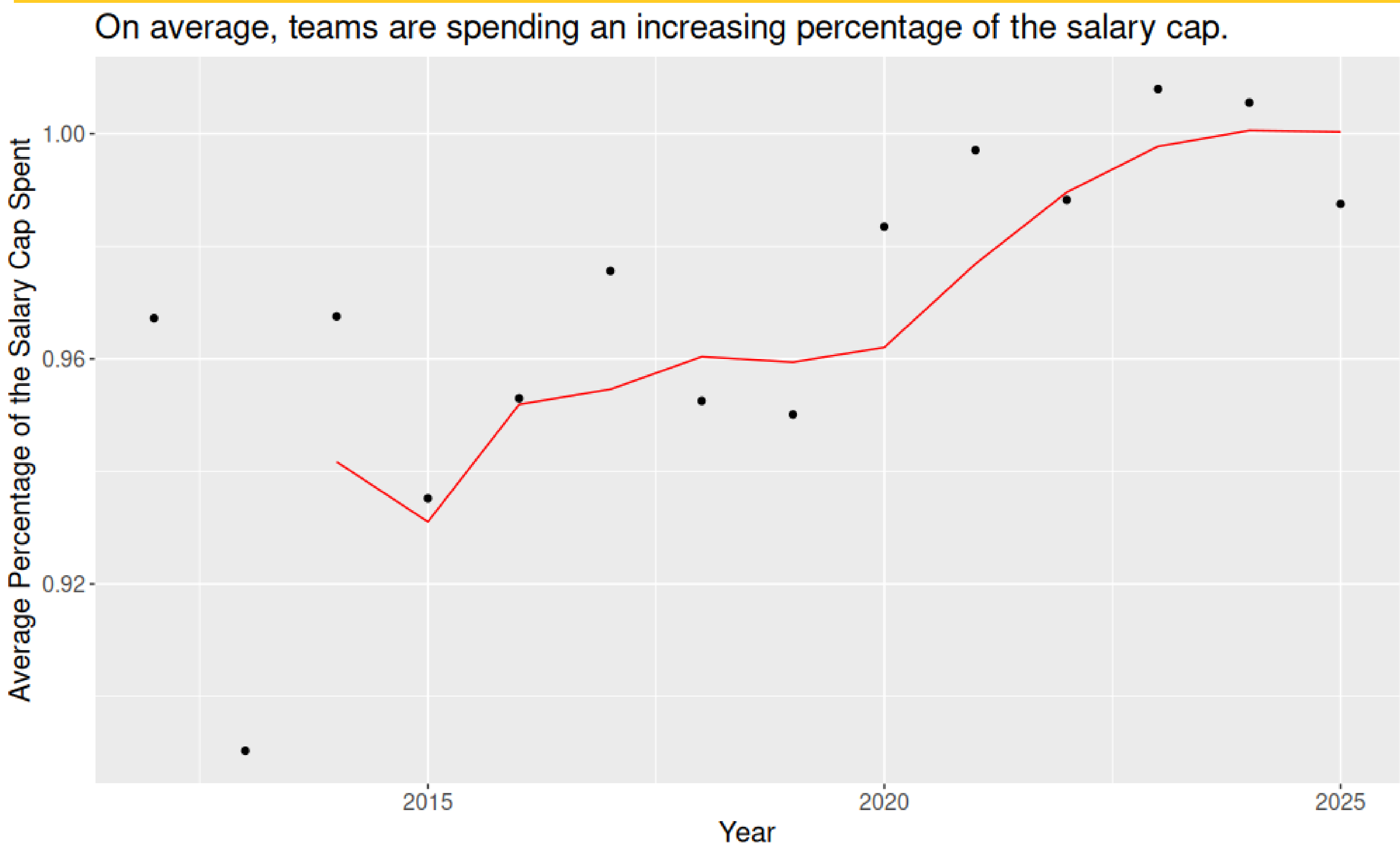
Moving Forward: NHL Forwards Provide More Value than Defensemen



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Background

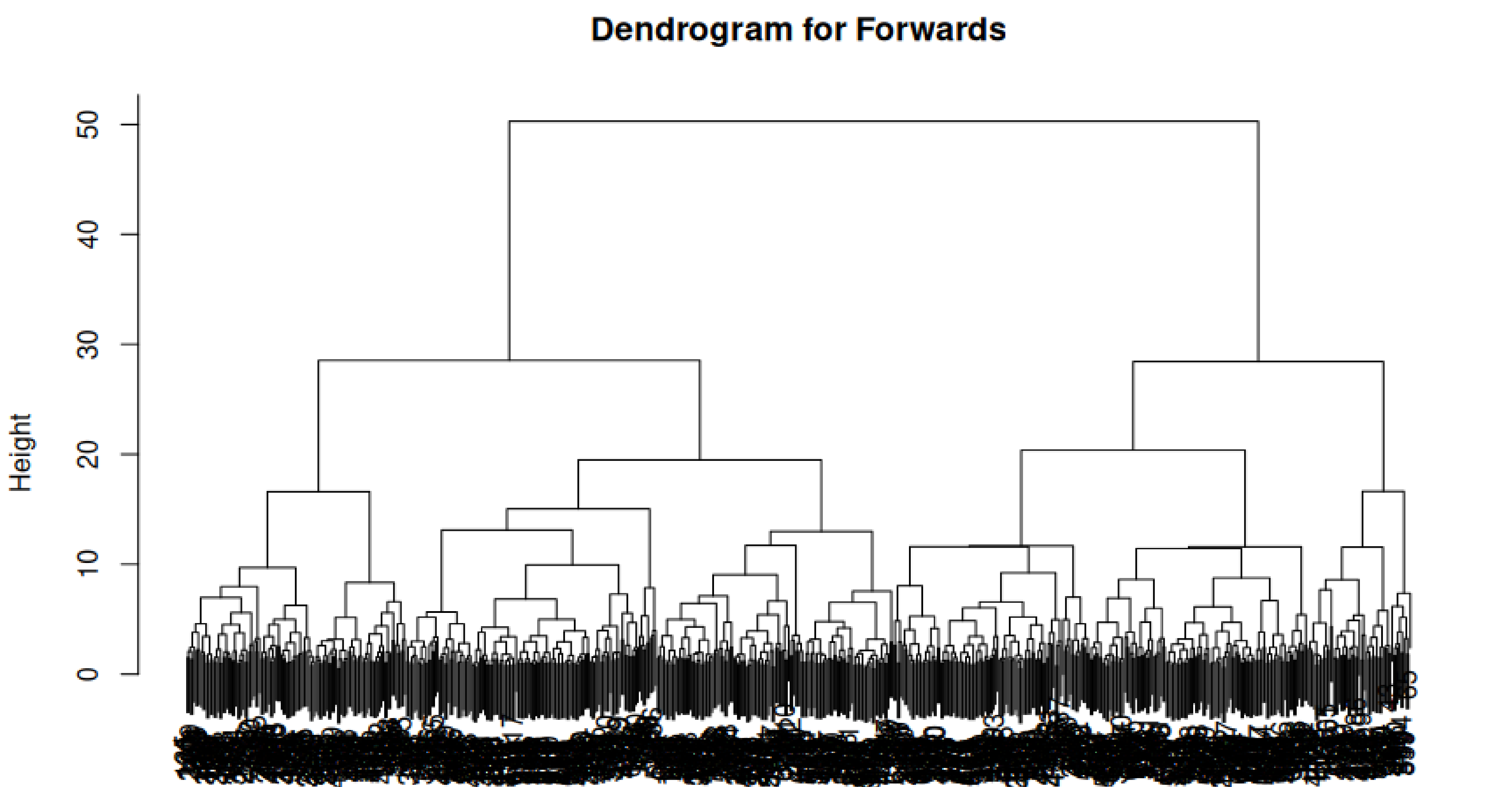


- Finding good value players is becoming increasingly important.
- Making and succeeding in the playoffs is a large contributor to a team's financial success, as teams can bring in upwards of \$10,000,000 for each playoff game.¹
- Hockey teams need a variety of players with different roles on the team.²
- Players are typically classified into different archetypes to help fit these roles.
- There is a lack of work on how many players of each archetype a team should ice to maximize their chances of making and succeeding in the playoffs.

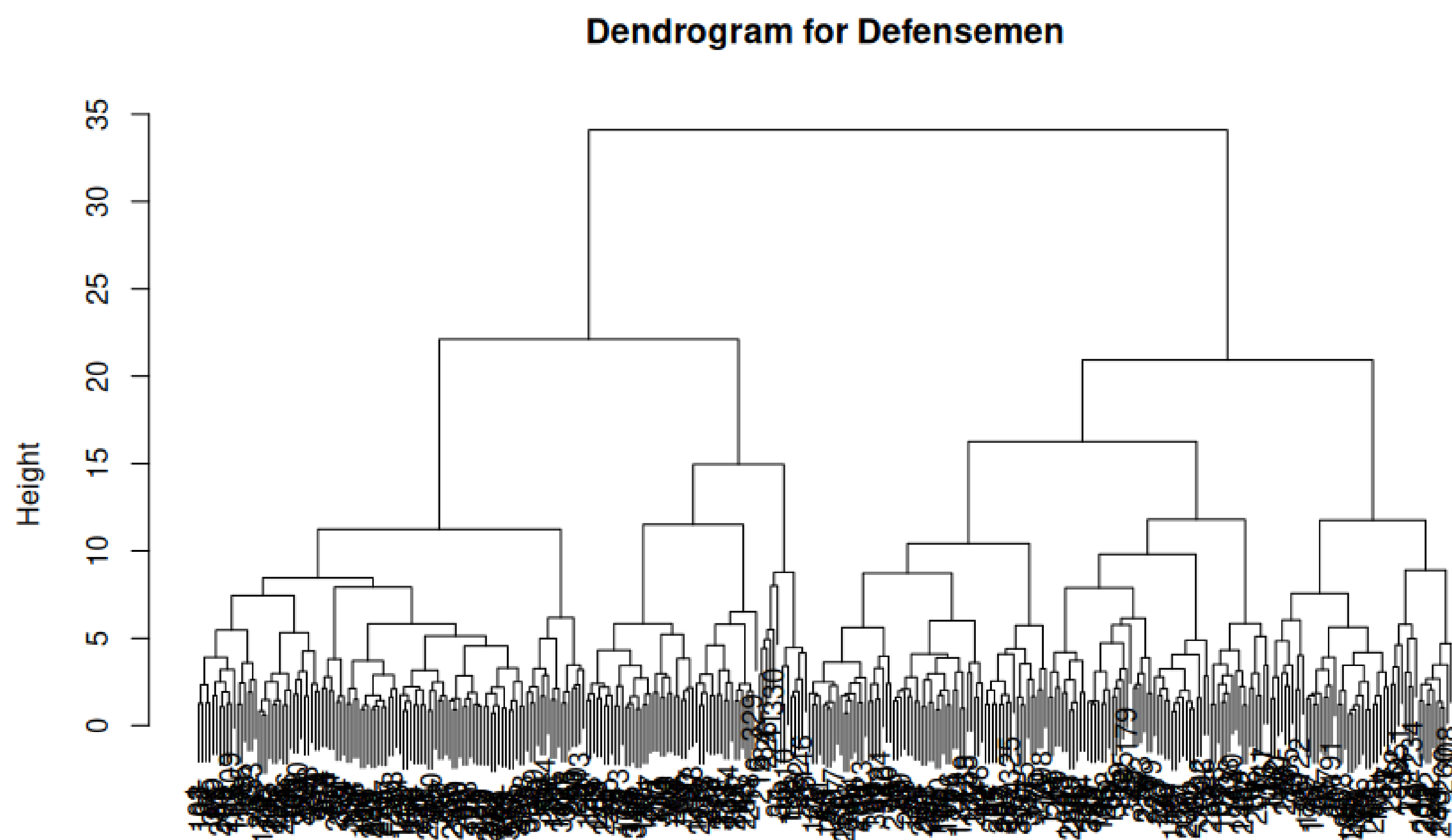
Methods

- Agglomerative cluster analysis was run on normalized player data for each position to mathematically cluster the players.
- xGoals data was obtained and merged with player shift data to determine the net impact of every shift.
- Clusters were ranked by each statistic.
- Each cluster was given a name using prior hockey knowledge, the ranks, and the summary statistics.
- Ridge-regularized weighted least squares regression was performed to model the xGoals from every shift using the player categories on the ice for that shift.
 - This follows the regularized adjusted plus minus (RAPM) method developed by Brian Macdonald³
 - The coefficients for each cluster represent the relative value of players in that cluster.

Results



Forward Cluster Ranks					Defensemen Cluster Ranks				
Stats (per 60)	Cluster Rank				Stats (per 60)	Cluster Rank			
Goals	1	3	2	4	Goals	4	3	2	1
Primary Assists	1	3	4	2	Primary Assists	4	3	1	2
Secondary Assists	3	1	4	2	Secondary Assists	4	3	1	2
Shot Attempts	1	3	4	2	Shot Attempts	4	2	3	1
Penalty Minutes	4	2	1	3	Penalty Minutes	2	1	4	3
Hits	4	2	1	3	Hits	2	1	3	4
Giveaways	1	3	4	2	Giveaways	3	4	2	1
Takeaways	1	3	2	4	Takeaways	4	3	2	1
Rebounds	1	3	4	2	Rebounds	4	3	2	1



Conclusions

- Four clusters were chosen for both forwards and defensemen.
- All four forward clusters had positive value, while all four defensemen clusters had negative value.
- The cluster of top line forwards had the greatest value, while middle six offensive forwards had the least value among forwards.
- Top pairing puck moving defensemen had the highest value among defensemen, while physical defensemen had the least value overall.
- The results suggest that teams should prioritize having forwards on the ice. More research is recommended to verify if the typical three forward two defensemen deployment is indeed the most optimal.

References

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Data Sources

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Cluster	Classification	xGF/60	xGA/60	xGD/60
F 1	Top Line Offense	0.631	-0.124	0.755
F 2	Power Forwards	0.448	-0.167	0.615
F 3	Middle Six Offense	0.464	-0.107	0.571
F 4	Bottom Six Grinders	0.467	-0.166	0.633
D 1	Bottom Four Shutdown Defensemen	-0.519	0.171	-0.690
D 2	Physical Defensemen	-0.513	0.210	-0.723
D 3	Bottom Four Puck Movers	-0.431	0.204	-0.635
D 4	Top Pairing Puck Movers	-0.289	0.124	-0.413